

LOGIC RELATIVITY — DARK CONTINUUM (LR-DC)

Physics Theory, Ontology, Simulation and Numerical-Validation Repository Dossier

Repository-ready documentation for the GitHub project

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Purpose of this document

This PDF is the repository dossier requested for the LR-DC project. It gives a single, ordered physics-level description of the ontology, mathematical framework, numerical implementation programme, simulation evidence, observational tests, and the current state of unresolved quantities. It is designed so that the theory and its numerical evidence are presented together rather than being scattered among separate folders after extraction.

Important distinction: a PDF can contain the complete documented repository description, equations, numerical tables, test ledger, and reproducibility map, but it cannot itself replace executable source code, raw simulation arrays, or machine-readable CSV files. The GitHub repository should therefore contain the actual code and data; this PDF should be the master human-readable map of what those files mean and how they connect.

1. Physics overview

LR-DC is a four-dimensional relativistic dark-continuum theory formulated on an Einstein spacetime. The dark sector is represented by one conserved, universe-filling continuum. Its microscopic degrees of freedom are three comoving scalar material coordinates. The continuum is intended to have different macroscopic regimes rather than being split into separate dark-matter and dark-energy substances.

In the clustered regime the continuum behaves as cold matter and supplies the dark-matter-like gravitational mass distribution of galaxies and larger structures. In a diffuse expanding regime the same ontology is intended to produce a dark-energy-like effective stress. The diffuse branch is a derivational objective rather than a quantity inserted into the theory by definition.

Ordinary baryonic matter couples minimally to the metric. There is no fifth dimension and no direct baryon-continuum fifth force. The physical feedback chain is:

baryonic and dark stress-energy $\rightarrow$ Einstein geometry $\rightarrow$ continuum displacement/deformation $\rightarrow$ dark-continuum stress-energy $\rightarrow$ Einstein geometry

2. Core ontology

- Spacetime is four-dimensional and described by the Einstein metric $g_{\mu\nu}$.
- The dark continuum is represented microscopically by three scalar material coordinates $\Phi^A(x)$, $A=1,2,3$.
- The continuum is conserved through its material current.
- Baryons interact with the continuum only through gravity and the metric.
- The continuum is not electromagnetic matter and is not a new particle species in the present ontology.
- The same continuum can form clustered cold structures or remain diffuse and cosmologically smooth.
- The cosmological constant is retained until a derived diffuse-DC branch demonstrably replaces it.
- The galaxy-scale infrared response is an effective collective description whose microscopic derivation remains a specific calculation.

3. Fundamental action

$$\begin{aligned} S &= S_{EH} + S_b + S_{DC} + S_{\Lambda} \\ S_{EH} &= (c^3 / 16 \pi G) \int d^4x \sqrt{-g} R \\ S_{DC} &= - \int d^4x \sqrt{-g} \epsilon_{DC}(b, Q) \\ \epsilon_{DC} &= \rho_0 c^2 b + W_{def}(Q) + W_P(b) \end{aligned}$$

The Einstein-Hilbert sector is retained. The continuum contribution is varied with respect to the metric and material coordinates to obtain its stress-energy and continuum equations of motion.

4. Material geometry and conservation

$$\begin{aligned} B^A(B) &= g^{\mu\nu} \partial_\mu \Phi^A \partial_\nu \Phi^B \\ C^A_B &= \gamma_{BC} B^A(C) \\ b &= \sqrt{\det C} \\ J^\mu &= (1/3!) \epsilon^{\mu\nu\rho\sigma} \epsilon_{ABC} \\ &\quad \times \partial_\nu \Phi^A \partial_\rho \Phi^B \partial_\sigma \Phi^C \\ \nabla_\mu J^\mu &= 0 \\ J^\mu &= b u^\mu \end{aligned}$$

The determinant b measures volume/compression of the material configuration. In a homogeneous FLRW configuration $b=a^3$.

5. Objective deformation

$$\begin{aligned} \bar{C}^A_B &= b^{(-2/3)} C^A_B \\ S &= (1/2) \ln(\bar{C}) \end{aligned}$$

$$Q = \text{Tr}(S^2) \geq 0$$

$$R = \text{Tr}(S^3)$$

$$D^2 = Q + \beta_R |R|^{2/3}, \text{ with } \beta_R = 0 \text{ in the minimal branch}$$

The separation between b and Q is central: isotropic compression can change b while leaving Q equal to zero. Q therefore represents shape/deviatoric deformation rather than ordinary cosmological dilution.

6. Constitutive sector

$$W_{\text{def}}(Q) = \mu_0 q_0^2 [\sqrt{1 + Q/q_0^2} - 1]$$

$$\mu_0 = 4 \times 10^{-16}$$

$$q_0 = 10^{-3}$$

$$\rho_P = c^5/(\hbar G^2)$$

W_P is a positive Planck-density regulator

The UV regulator is a regulator only. It is not used as an automatic bounce mechanism. Several earlier branches that attempted to use the regulator as a dynamical bounce were removed from the working theory.

7. Homogeneous cosmology

$$\begin{aligned}\text{FLRW: } b &= a^{-3}, \quad Q = 0 \\ \rho_{\text{DC}} &= \rho_{\text{DC},0} a^{-3} \\ p_{\text{DC}} &= 0 \\ H^2 &= (8 \pi G/3)(\rho_r + \rho_b + \rho_{\text{DC}}) + \Lambda c^2/3\end{aligned}$$

This is the analytically closed cold branch. It provides the cosmological dark-matter-like component without changing the Einstein gravitational field equations.

$$\begin{aligned}a_{\text{LR}} &= c \sqrt{G \rho_{\text{DC},0}} \\ a_{\text{LR}} &= c H_0 \sqrt{3 \Omega_{\text{DC}} / 8 \pi}\end{aligned}$$

For the representative normalization $\rho_{\text{DC},0}$ approximately $2.30 \times 10^{-27} \text{ kg m}^{-3}$, the resulting acceleration scale is approximately $1.175 \times 10^{-10} \text{ m s}^{-2}$.

8. Unified dark-continuum bookkeeping

$$\begin{aligned}\rho_{\text{DC}} &= \rho_c + \rho_d \\ p_{\text{DC}} &= p_d \\ \rho_{\text{dot}_c} + 3H \rho_c &= -\Gamma \\ \rho_{\text{dot}_d} + 3H(1+w_d)\rho_d &= \Gamma \\ \rho_{\text{dot}_{\text{DC}}} + 3H(\rho_{\text{DC}} + p_{\text{DC}}/c^2) &= 0\end{aligned}$$

This is the macroscopic target for a single ontology containing both clustered and diffuse regimes. The microscopic derivation of Γ and the effective diffuse equation of state is intentionally left as a later calculation.

9. Weak-field and phase-space representation

$$\begin{aligned}f(x, v, t) &= \int d^3q \rho_0(q) \delta^3[x - X(q, t)] \delta^3[v - \dot{X}(q, t)] \\ \rho_{\text{DC}} &= \int d^3v f \\ \partial_t f + v \cdot \nabla_x f - \nabla(\Psi) \cdot \nabla_v f &= 0 \\ \nabla^2 \Psi &= 4 \pi G (\rho_b + \rho_{\text{DC}})\end{aligned}$$

The phase-space representation permits multistreaming after shell crossing. This is important because a static Jeans equation cannot determine the nonlinear formation history by itself.

10. Spherical equations

$$\begin{aligned}r &= r(R) \\ \lambda_r &= r' \\ \lambda_t &= r/R \\ b &= R^2/(r'^2) \\ \delta &= \ln[r/(R r')] \\ Q &= (2/3) \delta^2 \\ p_r &= b W_b - W + (4/3) \delta W_Q \\ p_t &= b W_b - W - (2/3) \delta W_Q \\ p_r - p_t &= 2 \delta W_Q \\ dp_r/dr + (2/r)(p_r - p_t) &= -\rho_{\text{DC}} g \\ dg/dr + 2g/r &= 4 \pi G (\rho_b + \rho_{\text{DC}})\end{aligned}$$

11. Analytic similarity branch

$$\begin{aligned}r &= C R^n \\ b &\propto r^{(3/n - 3)} \\ \rho_{\text{DC}} &\propto r^{-2} \text{ requires } n = 3\end{aligned}$$

$r = C R^3$
 $b = 1/(3 C r^2)$
 ρ_{DC} proportional to r^{-2}
 $M_{DC}(<r)$ proportional to r
 g_{DC} proportional to r^{-1}
 $V_c^2 = \text{constant}$
 $\delta_\infty = -\ln(3)$
 $Q_\infty = (2/3)(\ln 3)^2$ approximately 0.80463

This is an intermediate similarity solution. A physical halo must eventually leave the r^{-2} regime and approach a finite excess over the cosmological DC background.

12. Galaxy scaling

$r_* = \sqrt{G M_b / a_{LR}}$
 $v_* = (G M_b a_{LR})^{1/4}$
 $M_{DC}(<r)/M_b$ approximately $A (r/r_*) + \dots$
 $V_f^2 = A \sqrt{G M_b a_{LR}}$
 $V_f^4 = A^2 G M_b a_{LR}$
 $\chi_{\text{form}} = A^2$

The fourth-power baryonic scaling follows from the similarity structure plus the universal acceleration scale. The coefficient must ultimately be predicted by formation dynamics rather than chosen separately for each galaxy.

13. Effective infrared collective sector

```
rho_DC = rho_mono - div(P_DC)
g = g_N + 4 pi G P
p = 4 pi G P/a_LR
W_IR = a_LR^2/(4 pi G) [p^2/2 + p^3/(3 chi) + alpha_IR p^5/(5 chi)]
g = g_N + a_LR p
g_N/a_LR = (p^2 + alpha_IR p^4)/chi
p^2 = 2x/[1 + sqrt(1 + 4 alpha_IR x)]
x = chi g_N/a_LR
```

The effective mapping is monotonic for the physical branch. At low acceleration it produces g approximately $\sqrt{\chi a_{LR} g_N}$. At high acceleration the response tends to the Newtonian limit.

$(g - g_N)/g_N$ approximately $(\chi/\alpha_{IR})^{1/4} (a_{LR}/g_N)^{3/4}$

14. Two-scale coarse graining

```
P_DC(k) = rho_DC [exp(-k^2 ell^2/2) - exp(-k^2 L^2/2)] xi(k)
F(k) = exp(-k^2 ell^2/2) - exp(-k^2 L^2/2)
```

The intended interpretation is that P_{DC} is a coarse-grained relative displacement between shorter- and longer-scale descriptions of the same continuum. This provides a route to an infrared collective response without introducing a fifth dimension or a new fundamental species.

15. Fold / multistream microscopic bridge

```
U(q;mu) = kappa q^3/3 - h mu q + m_5 q^5/5 + lambda_6 q^6/6
h mu = kappa s^2
Delta U = (4 kappa/3)s^3 - (2 m_5/5)s^5 + ...
p = 2 lambda_p s
C_3 = kappa/(6 lambda_p^3)
C_5 = -m_5/(80 lambda_p^5)
C_5/C_3 = -3 m_5/(40 kappa lambda_p^2)
m_5/(kappa lambda_p^2) = -8 alpha_IR
U_sssss/(U_sss lambda_p^2) = -96 alpha_IR
```

For α_{IR} approximately 0.3009683, the corrected target is $m_5/(\kappa \lambda_p^2)$ approximately -2.4077464. This algebra is the current microscopic-bridge target. The actual critical configuration, zero mode, stable-mode backreaction, and the physical polarization conversion factor λ_p still have to be obtained from the original continuum action.

16. SPARC galaxy rotation curves

The repository's galaxy programme uses the SPARC sample of 175 galaxies and compares observed rotation curves against baryonic gravity plus the DC response.

```
V_b^2 = V_gas^2 + 0.5 s_g V_disk^2 + 0.7 s_g V_bul^2
g_b = V_b^2/R
alpha_IR approximately 0.3009683
```

The frozen empirical calibration gives RMS approximately 10.55 km/s, MAE approximately 6.09 km/s, and reduced chi-squared approximately 4.45 for the population calibration. The median fitted stellar normalization is approximately 0.88 and the median galaxy response parameter is approximately 0.67. These are empirical fit descriptors, not fundamental constants.

The pointwise acceleration residual scatter is approximately 0.112 dex overall and approximately 0.103 dex in the deep regime. The fitted galaxy-to-galaxy response scatter is larger and is therefore not interpreted as fundamental random physics.

17. DDO 161

$R = 13.37$ kpc

V_{obs} approximately 66.1 km/s

V_b approximately 29.42 km/s

$V_{\text{DC,required}}$ approximately 59.19 km/s

$M_{\text{DC}}(<13.37 \text{ kpc})$ approximately 1.089×10^{10} solar masses

Diagnostic cored representation approximately 1.114×10^{10} solar masses

The difference is approximately 2.3 percent in that diagnostic representation. The test is not a complete validation because the baryonic input was simplified rather than being a full reproduction of the tabulated SPARC mass distribution.

18. Numerical formation programme

The repository contains multiple generations of particle-mesh and cosmological formation experiments. Earlier tests showed that naive cold collapse could produce multistreaming but did not reliably settle into the desired galactic state. Some earlier implementations also contained an acceleration/comoving-drift inconsistency. Those runs are retained as development history, not as final physical evidence.

The corrected cosmological equations are:

$$\nabla_x^2 \Phi = 4 \pi G a^2 \delta \rho$$

$$dv/dt + H v = -(1/a) \text{grad}_x \Phi + a_{\text{baryon,physical}}$$

$$dt = da/(aH)$$

$$dx/dt = v/a$$

$$S = \sqrt{[(d \ln \rho / d \ln r + 2)^2 + (d \ln V_c^2 / d \ln r)^2]}$$

Corrected equal-particle reruns with $N=2048$ and a 48^3 mesh, using seed amplitudes 0.3, 1 and 3, produced late-time regions close to ρ proportional to r^{-2} and V_c^2 approximately constant. The representative $a=1$ slopes were approximately:

Seed	$d \ln \rho / d \ln r$	$d \ln V_c^2 / d \ln r$	Interpretation
0.3	-2.0007	+0.0075	Close similarity
1.0	-2.0105	+0.0371	Close similarity
3.0	-2.0166	-0.0160	Close similarity

These results support the existence of a similarity-producing numerical basin in the tested setup. They do not by themselves establish the physical normalization, resolution convergence at fixed physical scales, or a relaxed equilibrium distribution.

19. Phase-space and Jeans test

$$2(1-\beta) \sigma_r^2 = v_f^2$$

$$\chi_{\text{Jeans}} = [2(1-\beta) \sigma_r^2 / v_*^2]^2$$

The latest formation runs can contain coherent radial streaming during active collapse. Consequently the stationary Jeans normalization cannot yet be used as if the system were already in equilibrium. This is a direct reason the phase-space normalization remains an unresolved numerical problem.

20. Outer halo transition

$$\rho_{\text{DC}} = \rho_{\text{DC},0} + \delta \rho$$

$$M_{\text{exc}} = 4 \pi \int \delta \rho r^2 dr < \infty$$

$$\epsilon = E_{\text{sep}} - E$$

$$\delta f \propto \epsilon^n \Theta(\epsilon)$$

$$\delta \rho \propto \psi^{(n+3/2)}$$

$$\text{Kepler region: } \delta \rho \propto r^{-(n+3/2)}$$

$$\text{Finite transition: } \delta \rho \propto (r_{\text{tr}} - r)^{(n+3/2)}$$

A lowered-isothermal mathematical control demonstrated a finite outer transition and an approximately r^{-2} region with nearly flat circular speed. This is a control of the phase-space mathematics rather than a proof that the microscopic DC action selects that exact distribution.

21. Lensing

$$\rho_P(r) = v_f^2 / (4 \pi G r^2)$$

$$\Sigma(b) = v_f^2 / (4 G b)$$

$$\kappa = \gamma_t = v_f^2 / (4 G b \Sigma_{\text{crit}})$$

$$\alpha_{\text{hat}} = 2 \pi v_f^2 / c^2$$

The lensing prescription is standard GR applied to the DC stress-energy. No separate photon-continuum interaction is introduced.

22. Bullet Cluster and merging systems

The ontology allows DC to behave as a collisionless gravitationally bound continuum while baryonic gas shocks and dissipates. A successful Bullet Cluster calculation would therefore require a self-consistent merger simulation followed by the same GR lensing calculation applied to the resulting total stress-energy. The repository has the conceptual and lensing framework, but a dedicated fully resolved 2D/3D merger validation is not yet represented as a completed result.

23. Solar-system / local-scale test

$$F(k) = \exp(-k^2 \ell^2/2) - \exp(-k^2 L^2/2)$$

For the illustrative coarse-graining values $L=10$ kpc and ℓ approximately 114 pc, the response is negligible at AU-scale wavenumbers. The same scale separation makes the correction extremely small at CMB wavenumbers around 0.2 Mpc^{-1} in the tested filter model.

This is a scale-suppression result. A complete post-Newtonian PPN parameter calculation is still a separate test.

24. CMB and early universe

$$\rho_{\text{DC}}/\rho_r \text{ at } z \text{ approximately } 10^9 = \text{approximately } 2.93 \times 10^{-6}$$

$$\Delta H/H \text{ approximately } 1.46 \times 10^{-6}$$

$$\Delta N_{\text{eff}} \text{ approximately } 2.18 \times 10^{-5}$$

$$\Delta Y_p \text{ approximately } 3 \times 10^{-7}$$

$$\Delta(D/H) \text{ approximately } 6 \times 10^{-11}$$

The estimates indicate that the cold DC component is effectively standard matter during BBN for the adopted normalization. The UV regulator is negligible at recombination and BBN.

```

Representative Planck-like background: H0 approximately 67.4 km/s/Mpc
Omega_m approximately 0.315
Omega_b h^2 approximately 0.02237
Omega_DC approximately 0.266
r_s approximately 144.27 Mpc
D_M approximately 13.86 Gpc
100 theta_* approximately 1.0406

```

The compressed CMB calculation used acoustic scale and distance information. It is useful as a background consistency test, but it is not a full TT/TE/EE Boltzmann-spectrum calculation. Individual CMB peak heights, TE/EE structure, damping-tail details, lensing reconstruction, and possible isocurvature modes require an actual perturbation-code implementation.

25. CMB test map

Observable/test	Current repository interpretation
Matter-radiation equality	Consistent with adopted cold-DC background normalization.
Sound horizon	Representative value calculated in the background model.
Acoustic angular scale	Compressed CMB test performed.
TT peak positions	Requires full Boltzmann spectrum for definitive test.
TT peak heights	Not yet a full LR-DC spectrum result.
TE	Full spectrum test required.
EE	Full spectrum test required.
Damping tail	Full spectrum test required.
CMB lensing	Full perturbation/lensing reconstruction required.
Isocurvature	Adiabatic branch used; relative entropy sector requires audit.

26. Test status language for the repository

For the public repository README, unresolved quantities should be described as UNKNOWN FOR NOW rather than being presented as failed or passed. This keeps the physics readable and avoids implying that an unperformed experiment has a result.

Area	Repository wording
Microscopic IR bridge	UNKNOWN FOR NOW — target is a first-principles reduction from the Φ^4 action.
Physical χ _form	UNKNOWN FOR NOW — formation simulations show the similarity structure but not yet its physical normalization.
Full CMB TT/TE/EE	UNKNOWN FOR NOW — compressed background tests exist; full Boltzmann spectra are pending.
Bullet Cluster	UNKNOWN FOR NOW — dedicated merger and lensing simulation required.
Full PPN test	UNKNOWN FOR NOW — scale suppression is established in the effective filter test, but PPN coefficients need calculation.
Nonlinear stability	UNKNOWN FOR NOW — local propagation estimates exist; coupled nonlinear proof remains.
Dark-energy derivation	UNKNOWN FOR NOW — macroscopic unification is formulated; microscopic diffuse branch is not yet derived.

27. What the GitHub repository should contain

The repository is the actual computational evidence base. The PDF is the human-readable master dossier. The GitHub repository should keep the following hierarchy so that simulation code, data, diagnostics and theory documents do not scatter into unrelated locations:

Directory	Purpose
README.md	Public physics overview, ontology, installation and reproduction map.
docs/	Theory manuscript, derivations, test reports, equations and audit documents.
theory/	Core symbolic definitions, action, constitutive laws, spherical reduction and cosmology.
formation/	Cosmological N-body/particle-mesh codes, initial conditions and formation diagnostics.
galaxies/	SPARC data processing, rotation-curve fitting and DDO 161 diagnostics.
cosmology/	Background, BBN, CMB compressed tests and future Boltzmann interface.
lensing/	Analytical lensing and future 2D/3D merger ray tracing.
solar_system/	Local weak-field, scale suppression and future PPN calculations.
microscopic_bridge/	Fold reduction, derivative machinery and eventual Φ^4 -to-W_IR calculation.
data/	Machine-readable input data and derived numerical tables.
results/	Figures, summary CSV files, convergence tables and reproducibility outputs.
tests/	Automated unit tests and numerical regression tests.
scripts/	One-command reproduction scripts for major experiments.

28. Numerical-validation contents

Yes: conceptually, the repository should be the complete computational record of the theory — not merely a PDF. It should contain the simulation programs, exact input parameters, initial conditions, raw/processed numerical outputs, CSV summaries, convergence diagnostics, figures, and scripts needed to regenerate the reported numbers.

- SPARC: 175-galaxy processing, baryonic decomposition, fitted parameters, residuals and population statistics.
- DDO 161: baryonic reconstruction, required DC mass, fitted diagnostic halo and rotation-curve comparison.
- Cosmological formation: corrected comoving equations, particle-mesh solver, seed scans, spin scans and radial diagnostics.
- Cold-sheet/multistream tests: folding, phase-space labels, polarization diagnostics and collapse behaviour.
- Outer transition: separatrix and lowered-isothermal control calculations.
- Microscopic bridge: fold algebra, stationary-branch checks, derivative tests and target coefficient relations.

- CMB/background: equality, sound horizon, distance, acoustic scale, BBN estimates and future full-spectrum interface.
- Lensing: spherical analytical predictions and future merger simulations.
- Solar system: filter-scale suppression and future PPN implementation.

29. Numerical validation philosophy

Every major numerical claim should have four associated files where practical: (1) the executable script, (2) the exact configuration or parameter file, (3) machine-readable output such as CSV/NPZ, and (4) a short human-readable report stating what was measured and what it does not prove. This prevents a result from becoming detached from the code that generated it.

30. Reproducibility checklist

- Record code version and commit hash.
- Record physical constants and unit conventions.
- Record particle number, mesh resolution, timestep and integration interval.
- Record initial seed and random-number seed.
- Record boundary conditions and softening prescription.
- Record baryonic mass model and assembly history.
- Record every fitted parameter separately from every fundamental parameter.
- Store radial profiles and convergence diagnostics in machine-readable form.
- Do not replace failed or unstable runs; retain them under a clearly labelled diagnostic/results history.
- Use the same cumulative-mass definition for every circular-speed calculation: $V_c^2 = G M(<r)/r$.

31. Physics test ledger

Test	Result represented in repository
4D ontology	Defined.
Three-scalar conserved continuum	Derived.
FLRW cold branch	Derived: ρ proportional to a^{-3} , $p=0$.
Spherical r^{-2} similarity	Derived analytically.
Flat circular-speed similarity	Derived from cumulative mass of the r^{-2} branch.
Outer finite-transition mathematics	Demonstrated by phase-space control.
SPARC 175 galaxies	Numerically calibrated.
DDO 161	Diagnostic numerical agreement; simplified baryonic input.
Corrected N-body formation	Similarity reproduced in tested seed basin.
Full formation convergence	UNKNOWN FOR NOW.
Physical χ_{form}	UNKNOWN FOR NOW.
Stationary Jeans normalization	UNKNOWN FOR NOW.
Microscopic IR derivation	UNKNOWN FOR NOW.
CMB background / acoustic scale	Compressed tests performed.
Full TT/TE/EE	UNKNOWN FOR NOW.
CMB lensing reconstruction	UNKNOWN FOR NOW.
BBN	Small estimated DC corrections.
Solar-system scale response	Strong scale suppression in effective filter model.
Full PPN	UNKNOWN FOR NOW.
Bullet Cluster	UNKNOWN FOR NOW.
Nonlinear coupled stability	UNKNOWN FOR NOW.

32. Removed branches

- Fifth dimension.
- Direct baryon-continuum fifth force.
- Automatic bounce from the positive Planck regulator.
- Negative UV energy.
- Primordial elastic/deformation waves inserted as a required ingredient.
- Arbitrary primordial ringing or oscillatory transfer functions.
- Phenomenological response-lag parameter.
- Fundamental hand-inserted MOND/RAR interpolation.
- Galaxy-specific NFW-like fitting parameters as fundamental physics.
- The earlier local 114-pc gradient completion as the final infrared law.

33. Repository README opening text

LOGIC RELATIVITY — DARK CONTINUUM (LR-DC)

LR-DC is a four-dimensional Einstein-gravity framework containing one conserved dark continuum represented microscopically by three comoving scalar material coordinates. The continuum behaves as a cold matter component when homogeneous and is hypothesized to develop collective infrared structure during nonlinear gravitational formation. The same ontology is intended to describe clustered dark matter and, in a separate diffuse regime, dark-energy-like stress.

The theory contains no fifth dimension and no direct baryon-continuum fifth force. Baryons couple minimally to the metric. The repository contains the mathematical formulation, spherical reduction, galaxy rotation-curve analysis, cosmological particle-mesh simulations, phase-space controls, lensing equations, CMB/background calculations, local-scale tests, and the microscopic-bridge programme.

The repository distinguishes established derivations, numerical demonstrations, empirical calibrations and quantities that are UNKNOWN FOR NOW. The purpose is to make the entire chain from ontology to simulation reproducible and auditable.

34. Final repository architecture

The clean way to prevent the problem shown in the uploaded GitHub screenshot is to keep one repository with one main branch and put the PDF in docs/. The simulations and numerical validation remain in their own named directories, while results are stored under results/ with the corresponding script path recorded in each report. Do not use separate Git branches as folders for simulation outputs.

Recommended top level:

```
LR-DC/  
  README.md  
  CITATION.cff  
  LICENSE  
  docs/  
  LR_DC_Master_Theory.pdf  
  derivations/  
  test_reports/  
  theory/  
  formation/  
  galaxies/  
  cosmology/  
  lensing/  
  solar_system/  
  microscopic_bridge/  
  data/  
  results/  
  tests/  
  scripts/
```

This structure keeps the repository itself intact. The PDF is the map; the code and numerical data are the evidence; the results directories contain the outputs generated by that code.

35. Closing physics statement

LR-DC is best understood as a single conserved relativistic continuum whose observable behaviour changes with configuration, compression, deformation, phase-space structure and coarse-graining. The central mathematical result is that a three-dimensional material continuum embedded in four-dimensional Einstein geometry can admit a cold homogeneous branch and a nonlinear spherical similarity branch with ρ proportional to r^{-2} and approximately flat circular speed. The numerical programme has begun to reproduce this similarity structure in corrected cosmological formation calculations.

The central unresolved physical problem is to derive the effective collective infrared response and its normalization from the microscopic Φ^4 action, and then demonstrate that the same theory survives the full cosmological, lensing, local-gravity and merger tests. Those quantities are recorded as UNKNOWN FOR NOW rather than being assigned unsupported outcomes.

This dossier is therefore intended to be the single human-readable physics reference for the LR-DC computational repository.